A Robust Unified Strategy for Maximum Torque per Ampere and Field Weakening in Permanent Magnet Synchronous Motor

Kunkun Zuo, Fengxiang Wang, *Senior Member, IEEE*, Zheng Li, Dongliang Ke, *Senior Member, IEEE*, Ralph Kennel, *Senior Member, IEEE*, and Marcelo Lobo Heldwein, *Senior Member, IEEE*

***Abstract*—The intricate control structure and demanding preparations undeniably make the high precision and performance realization of maximum torque per ampere (MTPA) and field-weakening (FW) strategies challenging in the permanent magnet synchronous motor (PMSM). This article proposes a straightforward yet potent strategy to surmount this challenge. By redefining the whole control process as a convex optimization and employing the indicator function to approximate constraints, the MTPA-FW control is seamlessly realized through the calculation of a single equation online, obviating the requirement for any supplementary operation. Meanwhile, a model correction method based on the moving horizon estimator (MHE) is designed to mitigate the impact of parameter variation and enhance the calculation accuracy. The experimental results have proved the superiority of the proposed strategy.**

***Index Terms*—Field weakening (FW), maximum torque per ampere (MTPA), permanent magnet synchronous motor (PMSM).**

Nomenclature

β	Current phase angle.
ω_e	Electrical rotation speed of rotor flux.
ψ_f	Permanent magnet flux linkage.
ψ_d	Flux linkage in the d-axis.
$i_{d/q}$	Stator current in the d/q-axis.
i_{amp}	Stator current amplitude.
$I_{\max}$	Maximum stator current amplitude.
$L_{d/q}$	Inductances in the d/q-axis.
p	Number of pole pairs.
R_s	Stator resistance.
T_e	Electromagnetic torque.
$T_{e\max}$	Maximum electromagnetic torque.
T_{ref}	Torque command.
T_s	Sampling time.
$u_{d/q}$	Voltage in the d/q-axis.
$U_{\max}$	Maximum voltage amplitude.

Manuscript received 22 August 2023; revised 23 December 2023; accepted 1 February 2024. Date of publication 6 February 2024; date of current version 20 March 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 52277070, in part by the China Postdoctoral Science Foundation under Grant 2023M730598, in part by the Science and Technology Program of Fujian Province under Grant 2022T3070, Grant 2021I0039, and Grant 2022HZ028010, in part by the Science and Technology Program of Fujian Province under Grant 2021T3062 and Grant 2022T3061, and in part by the Science and Technology Program of Quanzhou City under Grant 2023C001R, Grant 2023C002DW, and Grant 2021C020R. Recommended for publication by Associate Editor Ali N/A M. Bazzi. *(Corresponding author: Fengxiang Wang.)*

Kunkun Zuo is with the Chair of High-power Converter Systems, TUM School of Engineering Design, Technical University of Munich, 80333 Munich, Germany, and also with the Fujian Science and Technology Innovation Laboratory for Optoelectronic Information of China, Fuzhou 350108, China (e-mail: kunkun.zuo@tum.de).

Fengxiang Wang, Zheng Li, and Dongliang Ke are with the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Quanzhou 362200, China (e-mail: fengxiang.wang@fjirsm.ac.cn; lizheng4569@mails.ucas.ac.cn; dongliang.ke@fjirsm.ac.cn).

Ralph Kennel and Marcelo Lobo Heldwein are with the Chair of High-power Converter Systems, TUM School of Engineering Design, Technical University of Munich, 80333 Munich, Germany (e-mail: ralph.kennel@tum.de; marcelo.heldwein@tum.de).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TPEL.2024.3362869.

Digital Object Identifier 10.1109/TPEL.2024.3362869

I. Introduction

EXTENSIVE research has been dedicated to permanent magnet synchronous motor (PMSM) control strategies to generate appropriate current references that ensure that the drive system outputs the specified torque while realizing secondary targets, such as enhancing efficiency with maximum torque per ampere (MTPA) [1] and expanding the speed range with field weakening (FW) [2]. Considering the characteristics of electromagnetic torque, the former minimizes the current amplitude without influencing the torque output, thereby effectively reducing losses associated with the stator current. The latter concentrates on diminishing the back electromotive force by weakening the flux, consequently resulting in an extended speed operation range. From the perspective of control principles, published methods, including MTPA and FW, can be broadly categorized into two main classes: model-free method [3], [4], [5], [6], [7], [8], [9] and model-based method [10], [11], [12], [13], [14], [15], [16], [17], as shown in Table I.

The model-free category encompasses methods such as lookup table (LUT) [8], [9], signal injection [3], [4], virtual signal injection [5], [6], [18], reactive power [7], vector current control [21], [22], torque and flux control [23], and single current regulator [24], [25]. Among them, the LUT methods that adhere to the principles proposed in [26] and [27] have gained wide application due to their exceptional simplicity and reliability. These methods only need to search the preestablished table containing optimal current references, readily achieving the specified operating status. It is noteworthy that, considering

0885-8993 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF CONTROL METHODS FOR MTPA AND FW

	Model-free Methods				Model-based Methods	
Method	LUT	Signal Injection	Virtual Signal Injection	...	Offline Calculation	Online Calculation
Paper	[8], [9]	[3], [4]	[5], [6], [18]		[19]	[10]–[17], [20]
Principle	The optimal trajectories for different operation conditions mapped by experimental tests	Feedback control the current angle to maximize the output toque	Extract and control the partial derivative of torque with respect to the current angle	...	Solving a nonlinear optimization problem	Dividing the operation range into several segments and exploiting specific functions to calculate the optimal point
Prework	Massive work is necessary	Unnecessary	Depends on the robust strategy	...	Requiring a LUT composed of precalculated optimal points	Depends on the robust strategy
Robustness	Does not need parameters	Does not need parameters	Specific parameters are needed	...	Performance depends on parameters	Performance depends on parameters
Complexity	Very simple	Filters are needed	Filters are needed	...	Flowchart is needed	Flowchart is needed
Summary	1. Higher robustness 2. Lower dynamic caused by feedback structures and filters 3. Constrained by its foundational algorithm ...				1. Simpler control principle and higher dynamic 2. Control performance relies on parameter precision 3. Flowchart is needed to deal with constraints ...	

diverse operating conditions such as load torque, rotating speed, and temperature, the mapping for a precise and comprehensive LUT necessitates extensive testing efforts and specialized equipment. Other model-free methods are also constrained by implicit drawbacks inherited from their foundational algorithm. For instance, signal injection leads to heightened reliance on filters and negatively affects control performance, and the virtual signal injection may not yield precise results under low speed due to noise. In contrast, the model-based principle provides an attractive approach to circumvent these issues [28].

The model-based methods directly calculate the references by leveraging the mathematical model of the motor system. Although all the model-based methods share a common theoretical foundation, their implementation strategies diverge. Ge et al. [19] formulate a nonlinear optimization problem and an additional flowchart to derive current references for MTPA control. However, the difficulty of solving this problem necessitates the adoption of an LUT consisting of precomputed offline solutions. Therefore, in order to reduce the computational complexity, most papers select to divide the operation range into three [2] to five segments [11], or even more [20], and exploit specific solutions and algorithm-switching flowcharts to determine the optimal operation point within each segment. Furthermore, various solvers have been utilized, including Newton's method [20], the Levenberg–Marquardt algorithm [13], and Ferrari's method [16]. These solvers offer different advantages and tradeoffs, contributing to computational precision and convergence rate.

Similar to a coin having two sides, model-based methods, despite deriving benefits from intuitive realization principles and favorable dynamics, must concurrently contend with the repercussions of parameter changes on output performance. Therefore, to achieve the required control precision, it is essential to eliminate the adverse effects of parameter variation in model-based methods, especially concerning significant changes in inductances due to magnetic saturation [14], [19], [20]. Two strategies are typically adopted to tackle this issue and attain accurate calculation results: the LUT or predefined formulas [13], [29], [30], [31] and online parameter estimation [32], [33], [34], [35], [36]. Depending on the finite-element analysis or experimental characterization, the inductance variation with respect to the stator current is identified, and LUTs or formulas are constructed accordingly. On the other hand, identification methods have been adopted for parameter estimation, such as sliding-mode observer [32], extended Kalman filter [34], recursive least squares [35], model reference adaptive system [37], and signal injection techniques [36].

Based on the aforementioned discussion, model-based methods manifestly confront two pivotal limitations.

1) Though the operation region segmentation effectively mitigates the computational difficulty of the optimization problem, it simultaneously amplifies the complexity of the controller and poses challenges in practical implementation due to the introduction of algorithm-switching strategies and the utilization of specific solutions [17], [38], [39]. Meanwhile, to ensure stability and smoothness during region switching, the importance of the transition design between these solutions cannot be overstated.
2) The application of premade-based methods for parameter correction is laborious due to the need for specialized equipment or software, as well as an immense amount of prework involved. For online parameter estimation methods, careful attention must be given to crucial factors, such as the observer parameters [32], [37], convergence rate [35], filter dependence [36], and estimation accuracy under the saturated motor [1] during the controller's design.

Focusing on these problems, a robust and unified strategy is proposed, and its contributions can be summarized as follows.

1) The MTPA and FW operations undergo a comprehensive redefinition as an unconstrained minimization problem by formulating a purposeful convex optimization while incorporating the indicator function. This refined approach eliminates the necessity for supplementary algorithms or

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

decision processes, allowing the real-time calculation of a single equation to achieve MTPA and FW across the entire operation range.

2) This study proposes a moving horizon estimator for the PMSM to accurately estimate the inductances and iteratively correct the mathematical model during each control cycle. In the consideration of the properties of drive systems, an improved solver is devised to enhance the estimator's convergence rate and steady-state performance. The only parameter that requires adjustment is the estimation horizon.

The rest of this article is organized as follows. Section II presents the preliminary information, including the mathematical model of PMSM and an introduction to traditional MTPA and FW operations. In Section III, the proposed unified algorithm is elaborated. The model correction and the current controller are designed in Section IV. The experimental results are presented in Section V. Finally, Section VI concludes this article. In addition, the Appendix provides a concise analysis of maximum torque per volt (MTPV) utilizing the unified algorithm.

II. Preliminary

It is essential to emphasize that, constrained by the utilized PMSM platform, this article is mainly dedicated to the theoretical investigation and experimental verification of MTPA and FW. The MTPV operation can refer to the Appendix for further insights.

A. System Model

By disregarding irrelevant and negligible factors to facilitate the mathematical deduction, the voltage and electromagnetic torque mathematical model of PMSM in the d–q axis can be presented as follows [1], [2]:

$$\boldsymbol{u}_s = R_s\boldsymbol{i}_s + \boldsymbol{L}\frac{d\boldsymbol{i}_s}{dt} + \omega_e\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_p) \tag{1}$$

$$T_e = 1.5p\boldsymbol{i}_s^{\mathrm{T}}\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_p). \tag{2}$$

Here, $\boldsymbol{u}_s = [u_d,\ u_q]^{\mathrm{T}}$, $\boldsymbol{i}_s = [i_d,\ i_q]^{\mathrm{T}}$, $\boldsymbol{L} = [L_d,\ 0;\ 0,\ L_q]$, $\boldsymbol{j} = [0,\ -1;\ 1,\ 0]$, and $\boldsymbol{\psi}_p = [\psi_f,\ 0]^{\mathrm{T}}$.

Constraints stemming from the acceptable stator current and voltage within the drive system delimit the feasible operation region

$$\|\boldsymbol{i}_s\|_2 \le I_{\max} \tag{3}$$

$$\|\boldsymbol{u}_s\|_2 \le U_{\max}. \tag{4}$$

B. MTPA Operation

MTPA can be separately realized by manipulating the d–q axis current [8] or adjusting the current phase angle [5], [40]

$$i_d = \frac{\psi_f}{2(L_d - L_q)} - \sqrt{\frac{\psi_f^2}{4(L_d - L_q)^2} + i_q^2} \tag{5}$$

or

$$\beta = \sin^{-1}\left(\frac{-\psi_f + \sqrt{\psi_f^2 + 8i_{\mathrm{amp}}^2(L_q - L_d)^2}}{4i_{\mathrm{amp}}(L_q - L_d)}\right). \tag{6}$$

C. FW Operation

In the FW region, where the current limit remains inactive, the optimal point is situated at the intersection of the torque curve and the voltage limit [41]

$$i_d = \frac{\sqrt{\omega_e^{-2}U_{\max}^2 + (L_q i_q)^2} - \psi_f}{L_d}. \tag{7}$$

Conversely, when the current limit is activated, the optimal point is determined by simultaneously considering both the current and voltage limits

$$i_d = \frac{\sqrt{L_d^2\psi_f^2 - (L_d^2 - L_q^2)(\psi_f^2 + (L_q^2 - \omega_e^{-2})U_{\max}^2)} - L_d\psi_f}{L_d^2 - L_q^2}. \tag{8}$$

III. Unified Algorithm for MTPA and FW

In this study, the optimization problem for achieving a seamless integration of MTPA and FW is formulated as

$$\begin{aligned} \min_{\boldsymbol{i}_s}\quad & \|T_{\mathrm{ref}} - T_e\|_2^2 + \lambda\|\boldsymbol{i}_s\|_2^2 \\ \text{s.t.}\quad & T_e = 1.5p\boldsymbol{i}_s^{\mathrm{T}}\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\mathrm{p}}) \\ & \boldsymbol{u}_s = R_s\boldsymbol{i}_s + \boldsymbol{L}\frac{d\boldsymbol{i}_s}{dt} + \omega_e\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\mathrm{p}}) \\ & \boldsymbol{i}_s^{\mathrm{T}}\boldsymbol{i}_s \le I_{\max}^2 \\ & \boldsymbol{u}_s^{\mathrm{T}}\boldsymbol{u}_s \le U_{\max}^2\ . \end{aligned} \tag{9}$$

The traditional methods select the current amplitude as the sole term of the objective function and enforce $T_{\mathrm{ref}} = T_e$ to formulate an equality constraint. To guarantee the feasibility of the corresponding solution, such designs require a speed–torque (speed–current) LUT [1], [2], [19] or an additional inequality constraint [28] to confine the torque command within the permissible region (where the torque command T_{ref} is less than the maximum electromagnetic torque $T_{e\max}$).

In contrast, this work adopts the discrepancy between T_{ref} and T_e as the first term and chooses the squared 2-norm current, assigned a relatively small weighting factor ($\lambda \ll 1$), as the second term, thereby prioritizing the torque as the primary control objective. This structure avoids the requirement for auxiliary operations and ensures compatibility with the torque command that extends beyond the feasible region.

In (9), the elimination of equality constraints can be achieved through variable substitution. As for the intractable inequality constraints, which are the underlying reason for operation region segmentation adopted in most papers, this study first transforms them into piecewise functions that combine with the objective

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

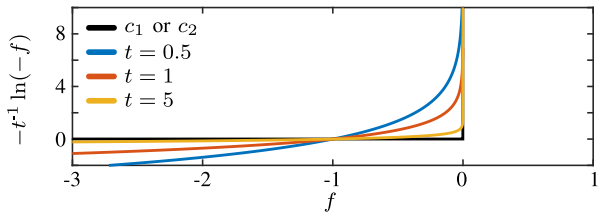


Fig. 1. Approximation performance of (11) with different t.

function

$$\min_{\boldsymbol{i}_s} \quad f_0 + \sum_{h=1}^{2} c_h \tag{10}$$

with

$$f_0 = \|T_{\text{ref}} - 1.5p\boldsymbol{i}_s^{\text{T}}\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}})\|_2^2 + \lambda\|\boldsymbol{i}_s\|_2^2$$

$$c_1 = \begin{cases} 0, & \boldsymbol{i}_s^{\text{T}}\boldsymbol{i}_s - I_{\max}^2 < 0 \\ \infty, & \boldsymbol{i}_s^{\text{T}}\boldsymbol{i}_s - I_{\max}^2 \geqslant 0 \end{cases}$$

$$c_2 = \begin{cases} 0, & (\omega_e\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}}))^{\text{T}}(\omega_e\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}})) - U_{\max}^2 < 0 \\ \infty, & (\omega_e\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}}))^{\text{T}}(\omega_e\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}})) - U_{\max}^2 \geqslant 0. \end{cases}$$

In c_2, the negligible voltage drop caused by resistance is disregarded [41], and the drive system is assumed to be in a steady state. Although (9) has been simplified through this equivalent transformation, it remains challenging to directly solve (10) because of the nondifferentiability of c_1 and c_2. This study employs (11) as the indicator function to formulate them approximately

$$\hat{c} = -t^{-1}\ln(-f). \tag{11}$$

Here, t determines the similarity between $\hat{c}$ and these piecewise functions, as plotted in Fig. 1. The parameter t should be configured with a value sufficiently large to ensure that (11) approximates the piecewise function satisfactorily. When $f < 0$, $\hat{c}$ exhibits a relatively small value. However, as f approaches 0, $\hat{c}$ experiences a rapid increase and ultimately becomes infinite when $f = 0$. These inequality constraints are consequently approximated with differentiable (11). It is noteworthy that, given the feature of (11), the logarithm is nonexistent in the following gradient calculation.

After eliminating the equality constraint and approximating the inequality constraints, the refined convex optimization problem for MTPA and FW is

$$\min_{\boldsymbol{i}_s} \quad f_0 - t^{-1}\sum_{h=1}^{2}\ln(-f_h) \tag{12}$$

with

$$f_1 = \boldsymbol{i}_s^{\text{T}}\boldsymbol{i}_s - I_{\max}^2$$

$$f_2 = (\omega_e\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}}))^{\text{T}}(\omega_e\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}})) - U_{\max}^2 .$$

This unconstrained minimization problem can be efficiently solved using Newton's method. Besides, to conserve computational resources, only one search step is calculated in each control cycle. The stator current reference in the whole speed region is

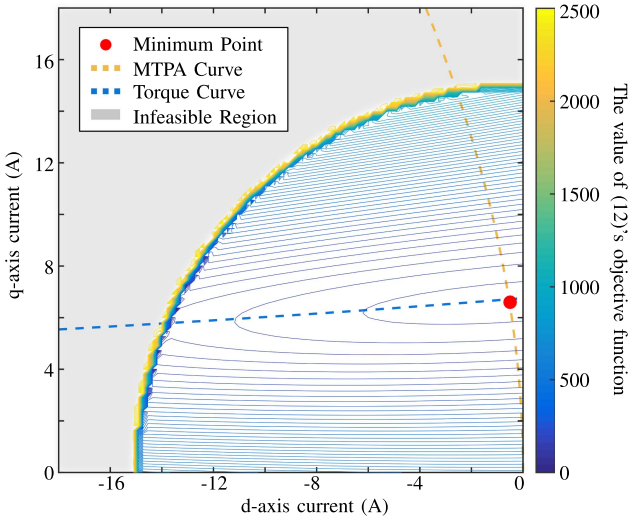


Fig. 2. Contour of (12)'s objective function in the MTPA region with an achievable torque command (MTPA operation—Case A).

$$\boldsymbol{i}_{\text{ref},k+1} = \boldsymbol{i}_{\text{ref},k} - \boldsymbol{H}_{\text{f},k}^{-1}\boldsymbol{J}_{\text{f},k} \tag{13}$$

with

$$\boldsymbol{H}_f = t\nabla^2 f_0 + \sum_{h=1}^{2} f_h^{-2}\nabla f_h \nabla f_h^{\text{T}} - \sum_{h=1}^{2} f_h^{-1}\nabla^2 f_h$$

$$\boldsymbol{J}_f = t\nabla f_0 - \sum_{h=1}^{2} f_h^{-1}\nabla f_h .$$

Here, $\boldsymbol{i}_{\text{ref},k} = [i_{\text{dref},k} \quad i_{\text{qref},k}]^{\text{T}}$. Simply calculating this equation online, the desired current reference in the d–q axis can be easily obtained. To underscore its efficacy theoretically, the subsequent subsections will elucidate the adaptive control principle of (12) in different scenarios.

A. MTPA Operation

1) Case A—Achievable Torque Command: When $T_{\text{ref}} < T_{e\max}$, the optimal point is located inside the feasible region and far away from the current and voltage limits. In this scenario, $-t^{-1}\ln(-f_1)$ and $-t^{-1}\ln(-f_2)$ are very close to zero, and (12) can be simplified to

$$\min_{\boldsymbol{i}_s} \quad f_0 . \tag{14}$$

This minimization problem can be regarded as the relaxation of the traditional MTPA problem (15)

$$\min_{\boldsymbol{i}_s} \quad \|\boldsymbol{i}_s\|_2$$
$$\text{s.t.} \quad T_{\text{ref}} = 1.5p\boldsymbol{i}_s^{\text{T}}\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}}). \tag{15}$$

The plotted contour of (12) (see Fig. 2) demonstrates that the calculated optimal point (minimum point) almost coincides with the intersection of the torque curve and the MTPA curve. The reason for the slight deviation is attributed to the design of

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

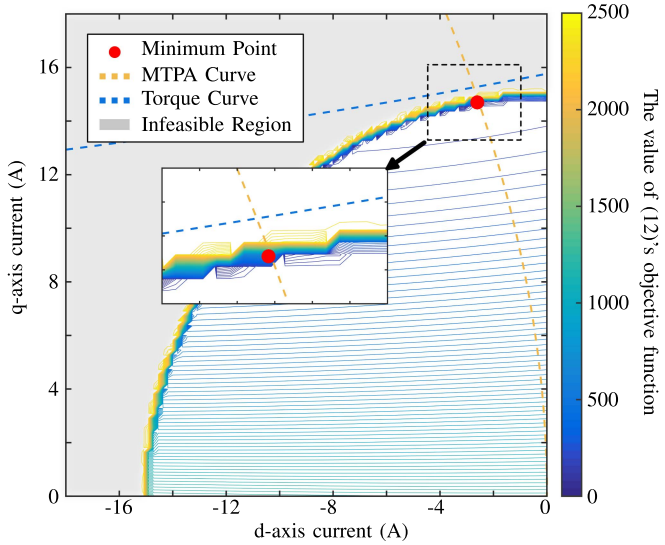


Fig. 3. Contour of (12)'s objective function in the MTPA region with an unachievable torque command (MTPA operation—Case B).

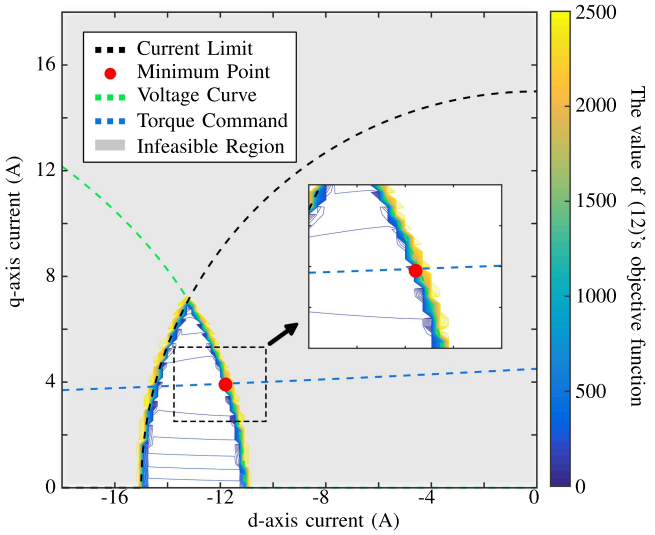


Fig. 4. Contour of (12)'s objective function in the FW region with an achievable torque command (FW operation—Case A).

f_0. Nevertheless, this phenomenon does not impact the control outcomes as the discrepancy is sufficiently small.

2) Case B—Unachievable Torque Command: If $T_{\text{ref}} \geq T_{\text{emax}}$ (see Fig. 3; the torque curve falls entirely in the unfeasible region), the optimal point should be the intersection of the MTPA curve and the current limit. In the unified algorithm, only the current constraint is activated and $-t^{-1}\ln(-f_2) \approx 0$; (12) can be rewritten as

$$\min_{\boldsymbol{i}_s} \quad f_0 - t^{-1}\ln(-f_1)\,. \tag{16}$$

The feature of $-t^{-1}\ln(-f_1)$ results in (12) changing slowly and gradually within the feasible region but experiencing a sharp increase in proximity to the current limit. To achieve the objective of minimizing (12), the optimal point must maximize the torque output within the current constraint. Consequently, the calculated minimum point corresponds to the intersection of the current limit and the MTPA curve, as illustrated in Fig. 3.

B. FW Operation

1) Case A—Achievable Torque Command: If $T_{\text{ref}} < T_{\text{emax}}$, the optimal operating point is determined by both the torque command and the voltage limit. Omitting the unactivated current limit ($-t^{-1}\ln(-f_1) \approx 0$), (12) is simplified into

$$\min_{\boldsymbol{i}_s} \quad f_0 - t^{-1}\ln(-f_2)\,. \tag{17}$$

The function of f_0 is to track the torque command and minimize the current, while $-t^{-1}\ln(-f_2)$ constrains the optimum point within the feasible area. Thus, the minimum point coincides with the intersection of the torque curve and the voltage limit, as shown in Fig. 4.

2) Case B—Unachievable Torque Command: Assuming that the torque command curve lies entirely within the infeasible region, the motor is required to output the maximum torque. To achieve this objective, (12) aims to minimize the distance between the torque command curve and the chosen point as much as possible. As a result, in Fig. 5, the calculated minimum point coincides with the intersection of the current and voltage limits.

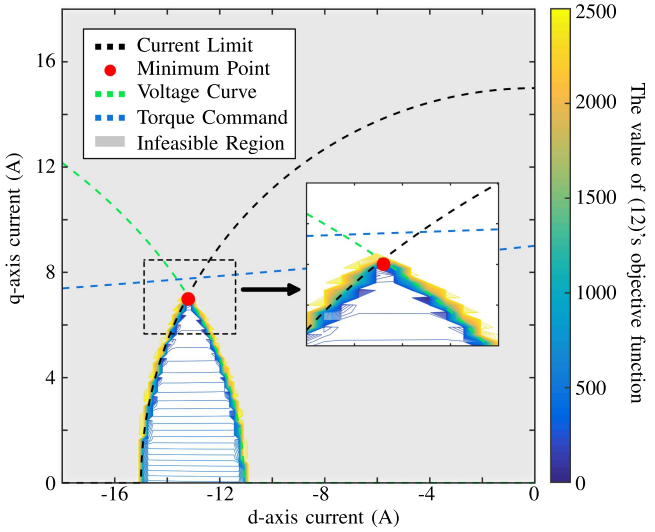


Fig. 5. Contour of (12)'s objective function in the FW region with an unachievable torque command (FW operation—Case B).

IV. Model Correction and Current Controller

The precision of the optimal point calculation and the tracking performance of the current loop are notably affected by the permanent magnetic flux linkage and inductances. While the former can be regarded as relatively constant within the allowable ranges of temperature and current amplitude, the latter exhibits fluctuations with variations in the stator current. Recognizing the imperative of parameter estimation in the unified method and model correction for the current controller, this work customizes

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

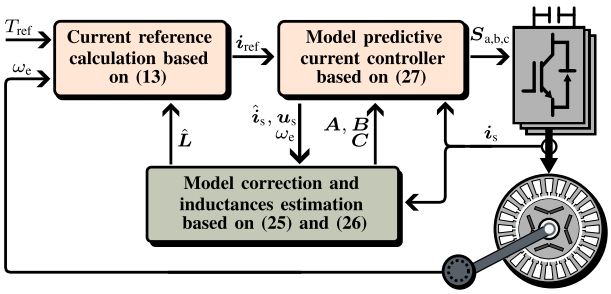


Fig. 6. Block diagram of the proposed strategy.

a moving horizon estimator and adopts the finite-control-set controller to fulfill these requirements. The control diagram is depicted in Fig. 6.

A. Model Correction

Based on (1), the stator current discrete time function in the practical PMSM drive system is

$$\boldsymbol{i}_{\text{s},k+1} = \boldsymbol{A}\boldsymbol{i}_{\text{s},k} + \boldsymbol{B}\boldsymbol{u}_{\text{s},k} + \boldsymbol{C}\omega_{e,k} + \boldsymbol{v}_k \tag{18}$$

with

$$\begin{aligned}\boldsymbol{A} &= \mathbb{I} - T_s R_s \boldsymbol{L}^{-1} - T_s \omega_{e,k} \boldsymbol{L}^{-1}\boldsymbol{j}\boldsymbol{L}\\ \boldsymbol{B} &= T_s \boldsymbol{L}^{-1}\\ \boldsymbol{C} &= -T_s \boldsymbol{L}^{-1}\boldsymbol{j}\boldsymbol{\psi}_{\text{p}}\,.\end{aligned}$$

Here, $\mathbb{I}$ is the identity matrix, and $\boldsymbol{v}_k = [v_{\text{d},k} \quad v_{\text{q},k}]^{\text{T}}$ is a vector composed of independent and identically distributed Gaussian noises with a mean of zero. Given parameter variations, a limited horizon optimization problem can be formulated as

$$\begin{aligned}\min_{\boldsymbol{A},\boldsymbol{B},\boldsymbol{C}} \quad & \boldsymbol{J} = \sum_{j=l}^{k} \|\boldsymbol{i}_{\text{s},j} - \hat{\boldsymbol{i}}_{\text{s},j}\|_2^2\\ \text{s.t.} \quad & \hat{\boldsymbol{i}}_{\text{s},k+1} = \boldsymbol{A}_k \boldsymbol{i}_{\text{s},k} + \boldsymbol{B}_k \boldsymbol{u}_{\text{s},k} + \boldsymbol{C}_k \omega_{e,k}\\ & \hat{\boldsymbol{i}}_s^{\text{T}} \hat{\boldsymbol{i}}_s - I_{\max}^2 \le 0\end{aligned} \tag{19}$$

where $\hat{}$ denotes the calculated variable, $l = k - h + 1$, and h is the estimation horizon. The objective of this optimization problem is to minimize estimated error within the estimation window by discarding the information outside the horizon and focusing on the last several control cycles. Replacing the predicted current in the objective function with the equality constraint renders the inequality constraint invalid, and (19) is transformed into

$$\min_{\boldsymbol{A},\boldsymbol{B},\boldsymbol{C}} \quad \|\mathcal{I}_{k+1} - \mathfrak{A}_k \mathcal{I}_k - \mathfrak{B}_k \mathcal{U}_k - \mathfrak{C}_k \mathcal{W}_k\|_2^2 \tag{20}$$

with

$$\begin{aligned}\mathcal{I}_k &= [\boldsymbol{i}_{\text{s},k} \quad \boldsymbol{i}_{\text{s},k-1} \quad \cdots \quad \boldsymbol{i}_{\text{s},l}]^{\text{T}}\\ \mathcal{U}_k &= [\boldsymbol{u}_{\text{s},k} \quad \boldsymbol{u}_{\text{s},k-1} \quad \cdots \quad \boldsymbol{u}_{\text{s},l}]^{\text{T}}\\ \mathcal{W}_k &= [\omega_{e,k} \quad \omega_{e,k-1} \quad \cdots \quad \omega_{e,l}]^{\text{T}}\\ \mathfrak{A}_k &= \text{diag}[\boldsymbol{A}_k \quad \boldsymbol{A}_{k-1} \quad \cdots \quad \boldsymbol{A}_l]\\ \mathfrak{B}_k &= \text{diag}[\boldsymbol{B}_k \quad \boldsymbol{B}_{k-1} \quad \cdots \quad \boldsymbol{B}_l]\\ \mathfrak{C}_k &= \text{diag}[\boldsymbol{C}_k \quad \boldsymbol{C}_{k-1} \quad \cdots \quad \boldsymbol{C}_l].\end{aligned}$$

This transformation turns (19) into an unconstrained optimization problem that can be promptly solved by the Gauss–Newton method

$$\delta\mathfrak{P} = -(\boldsymbol{\mathcal{J}}_{\mathfrak{P}}^{\text{T}} \boldsymbol{\mathcal{J}}_{\mathfrak{P}})^{-1} \boldsymbol{\mathcal{J}}_{\mathfrak{P}}^{\text{T}} (\mathcal{I}_{k+1} - \hat{\mathcal{I}}_{k+1}). \tag{21}$$

Here, $\mathfrak{P}$ denotes the matrix of $\mathfrak{A}$, $\mathfrak{B}$, or $\mathfrak{C}$, and $\boldsymbol{\mathcal{J}}_{\mathfrak{P}}$ denotes the Jacobi matrix of the objective function with respect to $\mathfrak{P}$. Despite the potential for significant and rapid parameter fluctuations induced by secondary effects such as the skin and proximity effect and magnetic saturation, these parameters can still be treated as constants when observed over several adjacent control cycles

$$\boldsymbol{P}_k \approx \boldsymbol{P}_{k-1} \approx \cdots \approx \boldsymbol{P}_l \tag{22}$$

where $\boldsymbol{P}$ denotes the parameter matrix of $\boldsymbol{A}, \boldsymbol{B}$, or $\boldsymbol{C}$. The search steps of the Gauss–Newton method from $t = l$ to $t = k$, thus, can be seen as sequential updates of parameter matrices. When the utilized parameter matrices deviate significantly from the actual values, a similar descent direction is maintained within the estimation horizon. Hence, we can deduce that

$$\begin{aligned}\|\sum_{j=l}^{k} \delta\boldsymbol{P}_j\|_2 &= \left\| \sum_{j=l}^{k} \left(\frac{\partial \boldsymbol{J}_j}{\partial \boldsymbol{P}_j}^{\text{T}} \frac{\partial \boldsymbol{J}_j}{\partial \boldsymbol{P}_j} \right)^{-1} \frac{\partial \boldsymbol{J}_j}{\partial \boldsymbol{P}_j}^{\text{T}} \Delta \boldsymbol{i}_{\text{s},j+1} \right\|_2\\ &> \|\delta \boldsymbol{P}_j\|_2\end{aligned} \tag{23}$$

where $\boldsymbol{J}$ is the objective function. After these parameter matrices approach the actual value, the descent direction should theoretically be close to zero. However, due to the influence of noise, the search step will exhibit fluctuations around the mean of the noise. Considering $E(\boldsymbol{v}_k) = 0$, the sum of search steps is

$$\begin{aligned}\|\sum_{j=l}^{k} \delta\boldsymbol{P}_j\|_2 &\approx \left\| \sum_{j=l}^{k} \left(\frac{\partial \boldsymbol{J}_j}{\partial \boldsymbol{P}_j}^{\text{T}} \frac{\partial \boldsymbol{J}_j}{\partial \boldsymbol{P}_j} \right)^{-1} \frac{\partial \boldsymbol{J}_j}{\partial \boldsymbol{P}_j}^{\text{T}} \boldsymbol{v}_j \right\|_2\\ &\le \|\delta \boldsymbol{P}_j\|_2\,.\end{aligned} \tag{24}$$

Hence, instead of relying on a single calculated search step, this approach adds up all the steps within the estimation window to enhance the convergence rate and diminish steady-state fluctuations. Omitting the tedious derivation, we can correct parameter matrices as

$$\begin{aligned}\boldsymbol{A}_{k+1} &= \boldsymbol{A}_k + \begin{bmatrix} \frac{\sum_{j=l}^{k} i_{\text{d},j} e_{\text{d},j}}{\sum_{j=l}^{k} \boldsymbol{i}_{\text{s},j}^2} & \frac{\sum_{j=l}^{k} i_{\text{q},j} e_{\text{d},j}}{\sum_{j=l}^{k} \boldsymbol{i}_{\text{s},j}^2} \\ \frac{\sum_{j=l}^{k} i_{\text{d},j} e_{\text{q},j}}{\sum_{j=l}^{k} \boldsymbol{i}_{\text{s},j}^2} & \frac{\sum_{j=l}^{k} i_{\text{q},j} e_{\text{q},j}}{\sum_{j=l}^{k} \boldsymbol{i}_{\text{s},j}^2} \end{bmatrix}\\ \boldsymbol{B}_{k+1} &= \boldsymbol{B}_k + \begin{bmatrix} \frac{\sum_{j=l}^{k} u_{\text{d},j} e_{\text{d},j}}{\sum_{j=l}^{k} \boldsymbol{u}_{\text{s},j}^2} & \frac{\sum_{j=l}^{k} u_{\text{q},j} e_{\text{d},j}}{\sum_{j=l}^{k} \boldsymbol{u}_{\text{s},j}^2} \\ \frac{\sum_{j=l}^{k} u_{\text{d},j} e_{\text{q},j}}{\sum_{j=l}^{k} \boldsymbol{u}_{\text{s},j}^2} & \frac{\sum_{j=l}^{k} u_{\text{q},j} e_{\text{q},j}}{\sum_{j=l}^{k} \boldsymbol{u}_{\text{s},j}^2} \end{bmatrix}\\ \boldsymbol{C}_{k+1} &= \boldsymbol{C}_k + \begin{bmatrix} \frac{\sum_{j=l}^{k} \omega_{e,j} e_{\text{d},j}}{\sum_{j=l}^{k} \omega_{e,j}^2} & \frac{\sum_{j=l}^{k} \omega_{e,j} e_{\text{q},j}}{\sum_{j=l}^{k} \omega_{e,j}^2} \end{bmatrix}^{\text{T}}.\end{aligned} \tag{25}$$

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.



Fig. 7. Photograph of the experimental platform.

Here, $\boldsymbol{e} = \boldsymbol{i}_{\text{ref}} - \boldsymbol{i}_s$. After this correction, inductances can be calculated as

$$\hat{\boldsymbol{L}} = T_s \boldsymbol{B}^{-1} . \tag{26}$$

B. Current Controller

Since this work does not involve the external signal injection, the parameter identification method relies entirely on the input and output signals of the controlled motor. Consequently, the choice of the current controller significantly influences the identification results. Following system identification theory, this work adopts a finite-control-set model-predictive current control method to track the current reference and provide ample excitation for the controlled system

$$\begin{aligned} \min_{\boldsymbol{u}_s} \quad & \|\boldsymbol{i}_{\text{ref},k+1} - \hat{\boldsymbol{i}}_{\text{s},k+1}\|_2^2 \\ \text{s.t.} \quad & \hat{\boldsymbol{i}}_{\text{s},k+1} = \boldsymbol{A}_k \boldsymbol{i}_{\text{s},k} + \boldsymbol{B}_k \boldsymbol{u}_{\text{s},k} + \boldsymbol{C}_k \omega_{e,k} \\ & \boldsymbol{u}_{\text{s}} \in \mathcal{U} \end{aligned} \tag{27}$$

where $\mathcal{U}$ is the action set and determined by the converter. The optimal action can be obtained by emulating all the feasible converter outputs.

V. Experimental Validation

The experimental platform adopts a limited-speed PMSM as the control objective and utilizes a control system based on a TMS320F28379D microprocessor. The overview picture of the platform is shown in Fig. 7, and the corresponding parameters are presented in Table II.

As obtaining the actual values of parameters is difficult, discrepancies between the utilized model and the practical system are common. Therefore, to simulate this condition, we conduct a start-up operation with different estimation horizons under parameter mismatches. In this simulation test, all the parameters of the PMSM are predefined with a 50% error, and the motor starts from zero speed to 500 r/min with a 10-N · m load torque. Here, the actual value of L_d is 10 mH, and L_q is 14.5 mH. The prediction error of the d–q axis current, which represents the accuracy of the entire mathematical model, and the estimated inductances are exhibited in Fig. 8. Overall, the prediction error swiftly decreases to zero within 0.004 s, while the estimated

TABLE II
Parameters

Symbol	Description	Nominal Value
R_s	Stator resistance	0.68 Ω
ψ_f	Permanent magnet flux	0.31 Wb
p	Polar pairs	4
P_n	Nominal power	4.8 kW
T_n	Nominal torque	24 N·m
ω_n	Nominal rotating speed	2000 min^{-1}
V_{dc}	DC-link voltage	520 V
I_{max}	Current limit	15 A
T_s	Sampling time	100 μs

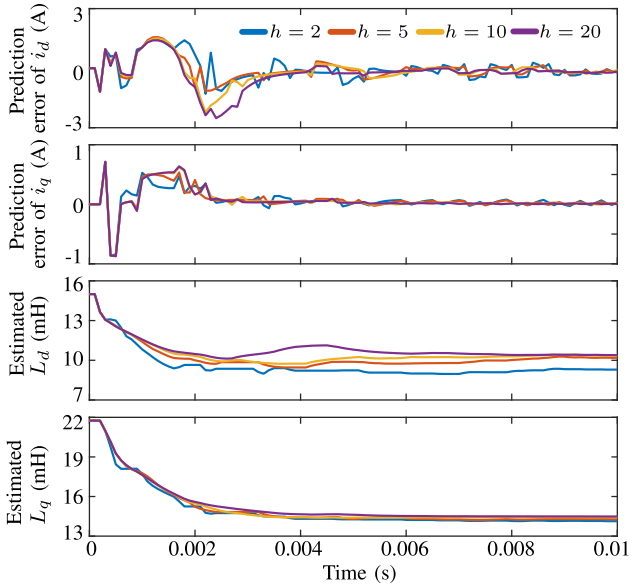


Fig. 8. Convergence performance of the proposed moving horizon estimator (MHE) with different estimation horizons under start-up operation and parameter mismatches.

inductances converge to real values. The estimation performance, including the convergence rate and precision, shows improvement with the increase of the horizon. The test results do not exhibit significant improvement when $h > 5$. Hence, we choose $h = 5$ as the length of the estimation window for the following tests.

While standstill identification methods have been widely adopted, their test results rely solely on stationary conditions and fail to adequately reveal parameter features during actual operations. In this work, we employ a load test presented in [20] to accurately identify PMSM's inductances under various conditions. The identification principle is grounded in the voltage model in the d–q axis. Under steady-state conditions, the derivative of current with respect to time can be ignored, simplifying the voltage equation to

$$\boldsymbol{u}_s = R_s \boldsymbol{i}_s + \omega_e \boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}}) . \tag{28}$$

Utilizing a power analyzer, we measure the phase voltage, fundamental components of the phase current, and the angle

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

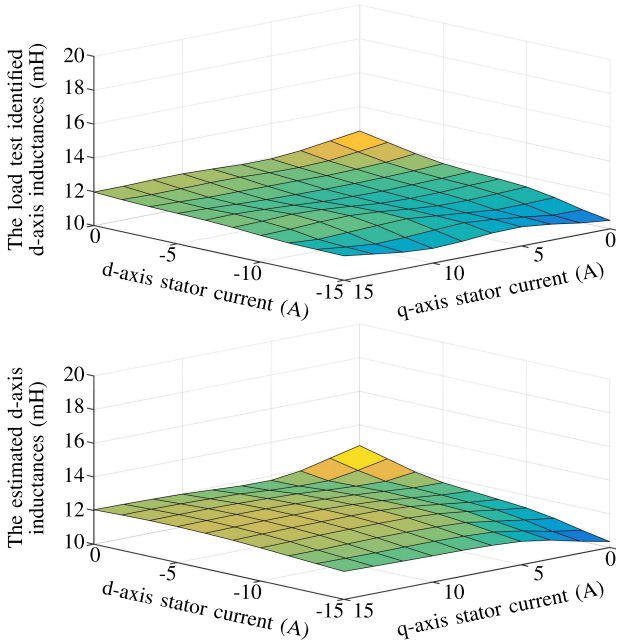


Fig. 9. d-axis inductances identified through the load test (upper) and estimated using MHE (lower).

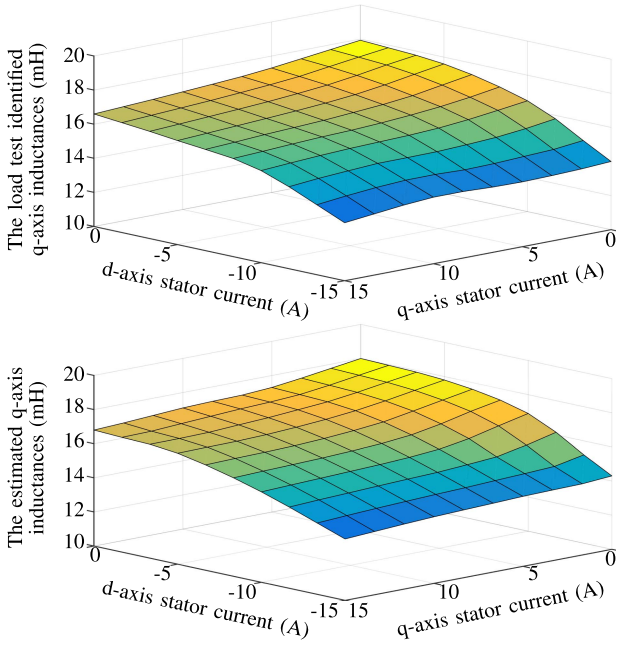


Fig. 10. q-axis inductances identified through the load test (upper) and estimated using MHE (lower).

between voltage and current. By controlling the PMSM under different scenarios, maps for inductances can be illustrated, as shown in Figs. 9 and 10. As the current amplitude increases, the inductances demonstrate a corresponding decrease, and, notably, the q-axis inductance exhibits a more pronounced nonlinearity compared with the d-axis inductance, showing a significant reduction both in identified data (from 17.9 to 13.4 mH) and estimated data (from 18.0 to 13.6 mH). The disparity between

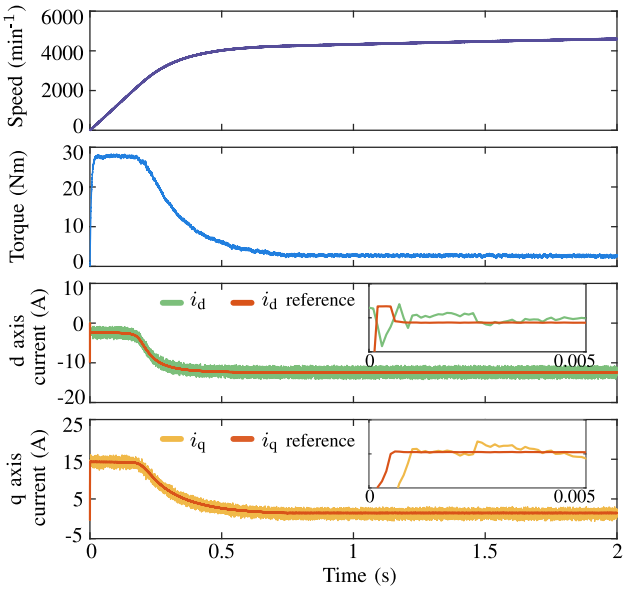


Fig. 11. Acceleration performance of the proposed method from zero to top speed. The torque reference is set to 40 N · m.

the identified and estimated data is relatively minor based on the comparative results. This outcome highlights the effectiveness of the designed method in achieving a high estimation accuracy.

An acceleration operation from zero to top speed provides a comprehensive evaluation of the proposed strategy, as shown in Fig. 11. In this test, the control algorithm must precisely deal with MTPA under the current constraint and FW under the current constraint while ensuring a seamless and smooth switch between them. We set the torque reference to a constant 40 N · m without load torque during this operation. Since the drive system cannot achieve the predetermined reference, in the beginning, the control algorithm generates the current reference to maximize the output torque as much as possible within the feasible region. As speed increases, the voltage limitation is activated, requiring the current reference to satisfy both the current and voltage constraints. According to Fig. 11, the rotating speed swiftly increases until the controlled motor enters the FW region. Subsequently, the generated torque slowly drops and so does the rate of increase in rotating speed. With the presence of friction, the output torque does not decrease to zero but reaches a certain point where the output torque equals the friction.

To provide more insights into control performance, the calculated current reference and sampled current under the acceleration operations are illustrated in the d–q axis, as shown in Fig. 12. Due to computational resource limitations, the solution to the formulated optimization problem is updated only once in every control cycle. The initial point is set to $i_d = 0$ and $i_q = 0$. It is worth noting that Newton's method requires several iterations to reach the minimum since the initial point is far from the optimum. We can observe that the current reference converges to the optimal point for MTPA operation after six iterations (0.0006 s). As the controlled system cannot reach the given torque reference throughout the process, the control

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

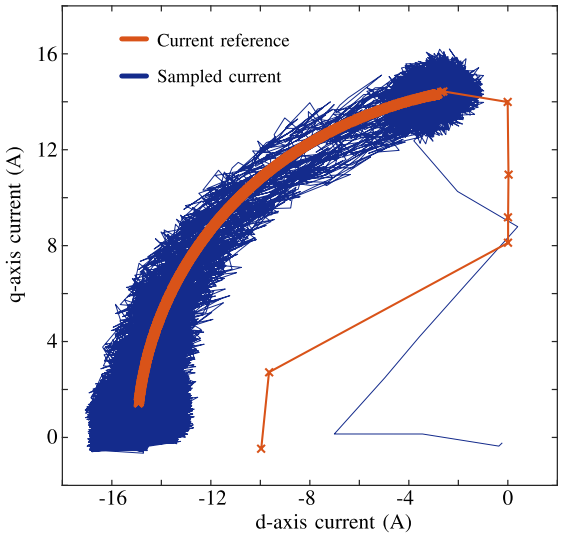


Fig. 12. Calculated current reference and the tracking performance of the proposed method under an acceleration operation.

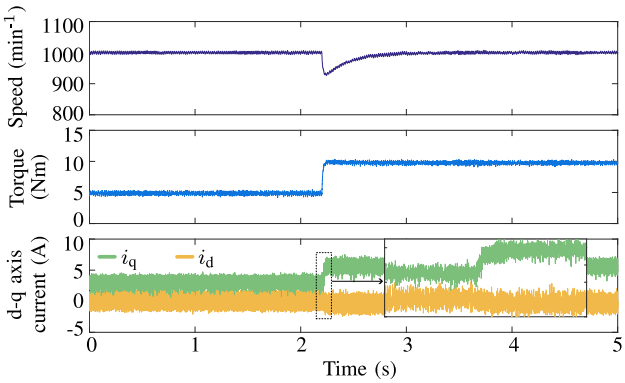


Fig. 13. Experimental results of load torque step change using the LUT-based controller.

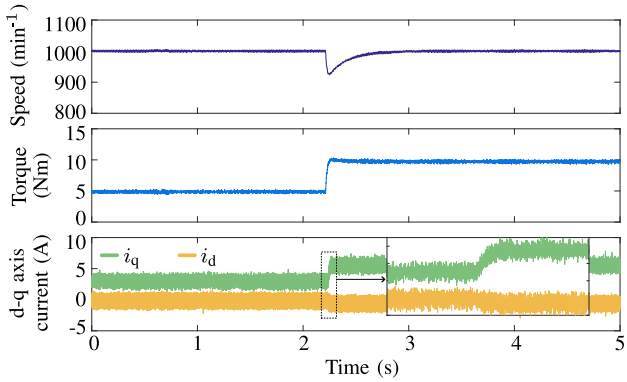


Fig. 14. Experimental results of load torque step change using the proposed controller.

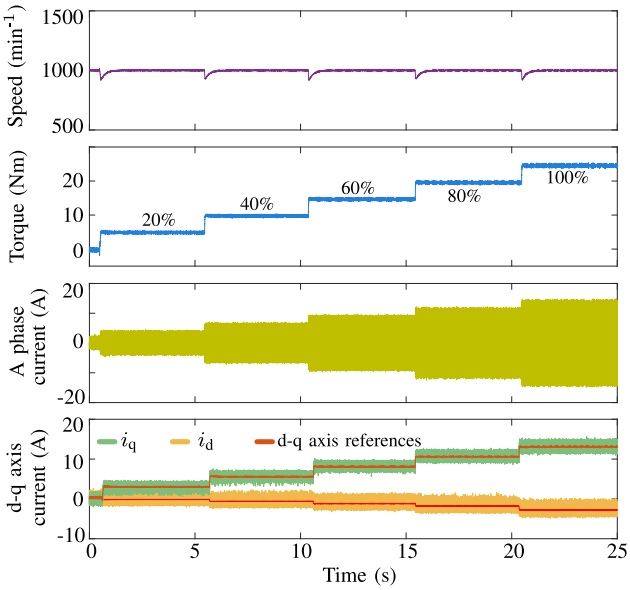


Fig. 15. Experimental results for MTPA operation using the LUT-based controller.

algorithm yields the stator current reference that maximizes electromagnetic torque under the present conditions. Consequently, the reference is distributed along the current limit circle (15 A). Because the current controller adopts a finite-control-set model-predictive control, the sampled current fluctuates significantly but remains closely distributed around the reference value.

The first comparative test involves using an LUT-based outer loop controller for a fair comparison, as illustrated in Figs. 13 and 14. During this test, the PMSM operates with a 1000-min^{-1} speed command and a step change in load torque from 5 to 10 N · m. The d–q axis currents undergo corresponding changes to prompt both the methods to meet the required speed and electromagnetic torque. The proposed method has minor search steps after achieving the near minimum point since the gradient is relatively small. Therefore, it demonstrates enhancements in steady condition, featuring slightly superior current quality and reduced torque fluctuations.

A sequence of step changes is introduced to incrementally raise the load torque from zero to full load, assessing the performance of both the LUT-based and proposed methods under MTPA operation, as depicted in Figs. 15 and 16. With the escalating load torque, the rotating speed correspondingly decreases. Subsequently, the torque regulator, a proportional–integral controller adopted in this work, generates the torque command to sustain the rotating speed at 1000 min^{-1}. The test outcomes reveal that both the methods demonstrate excellent dynamic and stable performance throughout the control process, characterized by the absence of delay, overshoot, and fluctuation. However, due to the utilization of a finite-control-set controller, notable current fluctuation results in a slight wave in torque and rotating speed.

To verify the control results under FW operation, Figs. 17 and 18 present tests that start at the nominal rotating speed (2000 min^{-1}) and gradually increase in steps of 250 min^{-1} until reaching the upper limit. It is crucial to note that the additional torque generated by friction correlates with the rotational speed, leading to an elevation in load torque. Furthermore, as illustrated

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

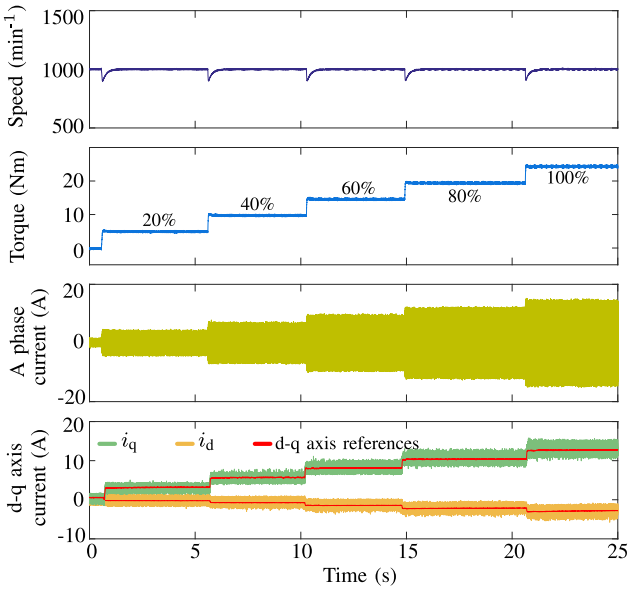


Fig. 16. Experimental results for MTPA operation using the proposed controller.

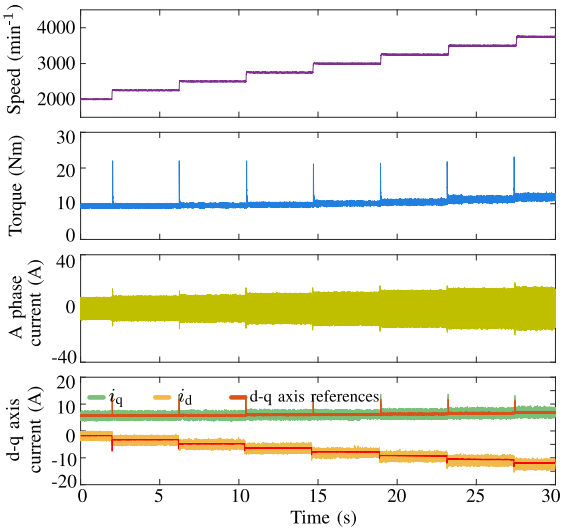


Fig. 17. Experimental results for FW operation using the LUT-based controller.

in Fig. 5, the disparity between the d-axis and q-axis inductances nonlinearly diminishes with an increase in current amplitude. Consequently, the q-axis current does not exhibit reduction during this test. To swiftly achieve the speed command, the torque regulator promptly yields a relatively higher torque. Due to the exclusive reliance on the preestablished table, the LUT-based method theoretically boasts the highest dynamic, consequently producing a small spike in d–q axis current. In contrast, the proposed method necessitates several control cycles to compute the optimal operation point under a step changing of torque reference, resulting in a more pronounced spike in its test results. Once the optimal point is attained, the proposed method demonstrates a steady-state performance similar to the LUT-based method.

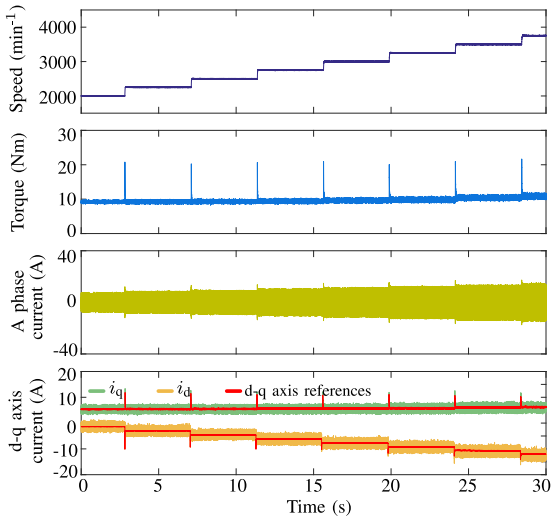


Fig. 18. Experimental results for FW operation using the proposed controller.

VI. Conclusion

In this article, we devise a unified algorithm that intricately merges MTPA and FW operations by meticulously formulating the optimization problem and addressing associated constraints. In comparison to traditional approaches, the proposed method stands out for its marked simplicity in both control principles and structure. Furthermore, a moving horizon estimator is thoughtfully designed to rectify parameter matrices and estimate inductances in real time, thereby ensuring the accuracy of the unified method and fortifying the robustness of the current control.

According to comparative tests, the proposed strategy demonstrates high accuracy in parameter estimation and delivers satisfactory control quality across the entire speed range. However, due to the limitation of computational resources, the unified method requires several control cycles to locate the optimal operation point, resulting in a slight lag behind the LUT-based method in dynamic performance. Nevertheless, considering the preparatory work involved in the LUT-based method, the proposed strategy boasts significant advantages in terms of rapid realization.

For future work, we aspire to explore a more comprehensive controller, enabling power optimization control across the full speed range.

Appendix A
MTPV Operation

In the MTPV region, the maximum output torque diminishes progressively as the increased rotating speed. Hence, the operation point is determined by the only intersection of the torque curve and voltage limit.

Classical model-based MTPV methods typically employ torque and voltage equations to compute this point [8], [20]

$$i_d = -\frac{\psi_f}{L_d} + \frac{-L_q\psi_f + \sqrt{(L_q\psi_f)^2 + 4L_q^2 i_q^2 (L_d - L_q)^2}}{2L_d(L_d - L_q)} \tag{29}$$

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

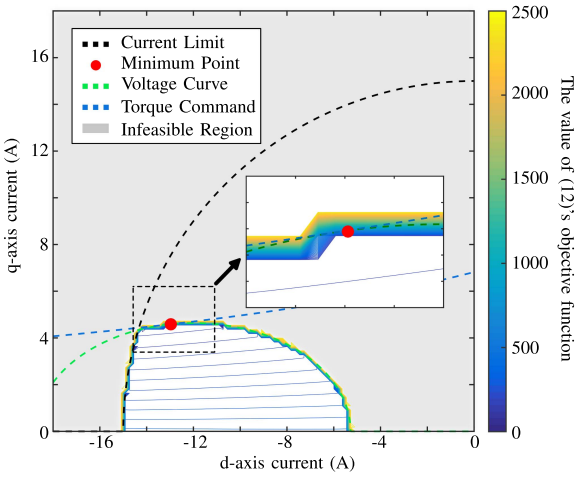


Fig. 19. Contour of (12)'s objective function in the MTPV region.

or

$$\psi_d = \frac{-L_q\psi_f + \sqrt{(L_q\psi_f)^2 + 8U_{\max}^2(L_d - L_q)^2/\omega_e^2}}{4(L_d - L_q)}. \quad (30)$$

In the formulated convex optimization, the current term within f_0 exhibits negligible influence on the determination of the optimal point in the MTPV region. In addition, the inactivated current constraint results in $-t^{-1}\ln(-f_1)$ approaching zero. Therefore, (12) can be approximated to

$$\min_{\boldsymbol{i}_s} \quad \|T_{\text{ref}} - 1.5p\boldsymbol{i}_s^{\text{T}}\boldsymbol{j}(\boldsymbol{L}\boldsymbol{i}_s + \boldsymbol{\psi}_{\text{p}})\|_2^2 - t^{-1}\ln(-f_2)\,. \quad (31)$$

Evidently, (31) is a simplified variant of the control principle of the traditional MTPV, and the illustration in Fig. 19 validates that the computed minimum point coincides with the intersection point.

REFERENCES

[1] A. Dianov, F. Tinazzi, S. Calligaro, and S. Bolognani, "Review and classification of MTPA control algorithms for synchronous motors," *IEEE Trans. Power Electron.*, vol. 37, no. 4, pp. 3990–4007, Apr. 2022.

[2] C. Miguel-Espinar, D. Heredero-Peris, R. Villafafila-Robles, and D. Montesinos-Miracle, "Review of flux-weakening algorithms to extend the speed range in electric vehicle applications with permanent magnet synchronous machines," *IEEE Access*, vol. 11, pp. 22961–22981, 2023.

[3] S. Bolognani, R. Petrella, A. Prearo, and L. Sgarbossa, "Automatic tracking of MTPA trajectory in IPM motor drives based on AC current injection," *IEEE Trans. Ind. Appl.*, vol. 47, no. 1, pp. 105–114, Jan./Feb. 2011.

[4] N. Bedetti, S. Calligaro, C. Olsen, and R. Petrella, "Automatic MTPA tracking in IPMSM drives: Loop dynamics, design, and auto-tuning," *IEEE Trans. Ind. Appl.*, vol. 53, no. 5, pp. 4547–4558, Sep./Oct. 2017.

[5] T. Sun, J. Wang, and X. Chen, "Maximum torque per ampere (MTPA) control for interior permanent magnet synchronous machine drives based on virtual signal injection," *IEEE Trans. Power Electron.*, vol. 30, no. 9, pp. 5036–5045, Sep. 2015.

[6] S. CallIgaro, D. Marzona, R. Petrella, and A. Shahdadi, "MTPA tracking algorithms for IPMSMs and synRMs: Accurate evaluation and adaptive tuning of real signal injection and virtual signal injection," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2020, pp. 6079–6086.

[7] J.-i. Itoh, Y. Nakajima, and M. Kato, "Maximum torque per ampere control method for IPM synchronous motor based on V/f control," in *Proc. IEEE 10th Int. Conf. Power Electron. Drive Syst.*, 2013, pp. 1322–1327.

[8] B. Cheng and T. R. Tesch, "Torque feedforward control technique for permanent-magnet synchronous motors," *IEEE Trans. Ind. Electron.*, vol. 57, no. 3, pp. 969–974, Mar. 2010.

[9] Y. Chen et al., "Improved flux-weakening control of IPMSMs based on torque feedforward technique," *IEEE Trans. Power Electron.*, vol. 33, no. 12, pp. 10970–10978, Dec. 2018.

[10] K. Choi, Y. Kim, K.-S. Kim, and S.-K. Kim, "Real-time optimal torque control of interior permanent magnet synchronous motors based on a numerical optimization technique," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 4, pp. 1815–1822, Jul. 2021.

[11] J. Lemmens, P. Vanassche, and J. Driesen, "PMSM drive current and voltage limiting as a constraint optimal control problem," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 3, no. 2, pp. 326–338, Jun. 2015.

[12] A. Consoli, G. Scelba, G. Scarcella, and M. Cacciato, "An effective energy-saving scalar control for industrial IPMSM drives," *IEEE Trans. Ind. Electron.*, vol. 60, no. 9, pp. 3658–3669, Sep. 2013.

[13] H.-S. Kim, Y. Lee, S.-K. Sul, J. Yu, and J. Oh, "Online MTPA control of IPMSM based on robust numerical optimization technique," *IEEE Trans. Ind. Appl.*, vol. 55, no. 4, pp. 3736–3746, Jul./Aug. 2019.

[14] Y. Lu, S. He, C. Li, H. Luo, H. Yang, and R. Zhao, "Online full-speed region control method of IPMSM drives considering cross-saturation inductances and stator resistance," *IEEE Trans. Transp. Electrific.*, vol. 9, no. 2, pp. 3164–3176, Jun. 2023.

[15] M. Preindl and S. Bolognani, "Model predictive direct torque control with finite control set for PMSM drive systems, Part 2: Field weakening operation," *IEEE Trans. Ind. Inform.*, vol. 9, no. 2, pp. 648–657, May 2013.

[16] S.-Y. Jung, J. Hong, and K. Nam, "Current minimizing torque control of the IPMSM using Ferrari's method," *IEEE Trans. Power Electron.*, vol. 28, no. 12, pp. 5603–5617, Dec. 2013.

[17] Z. Xia et al., "Computation-efficient online optimal tracking method for permanent magnet synchronous machine drives for MTPA and flux-weakening operations," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 5, pp. 5341–5353, Oct. 2021.

[18] Q. Chen, W. Zhao, G. Liu, and Z. Lin, "Extension of virtual-signal-injection-based MTPA control for five-phase IPMSM into fault-tolerant operation," *IEEE Trans. Ind. Electron.*, vol. 66, no. 2, pp. 944–955, Feb. 2019.

[19] H. Ge, Y. Miao, B. Bilgin, B. Nahid-Mobarakeh, and A. Emadi, "Speed range extended maximum torque per ampere control for PM drives considering inverter and motor nonlinearities," *IEEE Trans. Power Electron.*, vol. 32, no. 9, pp. 7151–7159, Sep. 2017.

[20] Z. Xia, S. Nalakath, R. Tarvirdilu-Asl, Y. Sun, J. Wiseman, and A. Emadi, "Online optimal tracking method for interior permanent magnet machines with improved MTPA and MTPV in whole speed and torque ranges," *IEEE Trans. Power Electron.*, vol. 35, no. 9, pp. 9753–9769, Sep. 2020.

[21] L. Sepulchre, M. Fadel, M. Pietrzak-David, and G. Porte, "MTPV flux-weakening strategy for PMSM high speed drive," *IEEE Trans. Ind. Appl.*, vol. 54, no. 6, pp. 6081–6089, Nov./Dec. 2018.

[22] D. Ding, G. Wang, N. Zhao, G. Zhang, and D. Xu, "Enhanced flux-weakening control method for reduced dc-link capacitance IPMSM drives," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 7788–7799, Aug. 2019.

[23] S. Ekanayake, R. Dutta, M. F. Rahman, and M. X. Bui, "Performances of a fractional-slot concentrated-winding permanent magnet synchronous machine under position sensorless control in deep flux-weakening region," *IEEE Trans. Ind. Appl.*, vol. 55, no. 6, pp. 5938–5946, Nov./Dec. 2019.

[24] Y. Zhang, M. K. Guven, M. K. Guven, S. Chi, and M. Illindala, "Experimental verification of deep field weakening operation of a 50-kW IPM machine by using single current regulator," *IEEE Trans. Ind. Appl.*, vol. 47, no. 1, pp. 128–133, Jan./Feb. 2011.

[25] L. Zhu, S. Xue, X. Wen, Y. Li, and L. Kong, "A new deep field-weakening strategy of IPM machines based on single current regulator and voltage angle control," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2010, pp. 1144–1149.

[26] S. Macminn and T. Jahns, "Control techniques for improved high-speed performance of interior PM synchronous motor drives," *IEEE Trans. Ind. Appl.*, vol. 27, no. 5, pp. 997–1004, Sep./Oct. 1991.

[27] S. Morimoto, M. Sanada, and Y. Takeda, "Wide-speed operation of interior permanent magnet synchronous motors with high-performance current regulator," *IEEE Trans. Ind. Appl.*, vol. 30, no. 4, pp. 920–926, Jul./Aug. 1994.

[28] M. Preindl and S. Bolognani, "Optimal state reference computation with constrained MTPA criterion for PM motor drives," *IEEE Trans. Power Electron.*, vol. 30, no. 8, pp. 4524–4535, Aug. 2015.

[29] W. Huang, Y. Zhang, X. Zhang, and G. Sun, "Accurate torque control of interior permanent magnet synchronous machine," *IEEE Trans. Energy Convers.*, vol. 29, no. 1, pp. 29–37, Mar. 2014.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.

[30] T. Inoue, Y. Inoue, S. Morimoto, and M. Sanada, "Mathematical model for MTPA control of permanent-magnet synchronous motor in stator flux linkage synchronous frame," *IEEE Trans. Ind. Appl.*, vol. 51, no. 5, pp. 3620–3628, Sep./Oct. 2015.
[31] Y. Miao, H. Ge, M. Preindl, J. Ye, B. Cheng, and A. Emadi, "MTPA fitting and torque estimation technique based on a new flux-linkage model for interior-permanent-magnet synchronous machines," *IEEE Trans. Ind. Appl.*, vol. 53, no. 6, pp. 5451–5460, Nov./Dec. 2017.
[32] Y. Han, C. Gong, G. Chen, Z. Ma, and S. Chen, "Robust MTPA control for novel EV-WFSMs based on pure SM observer based multistep inductance identification strategy," *IEEE Trans. Ind. Electron.*, vol. 69, no. 12, pp. 12390–12401, Dec. 2022.
[33] S. J. Underwood and I. Husain, "Online parameter estimation and adaptive control of permanent-magnet synchronous machines," *IEEE Trans. Ind. Electron.*, vol. 57, no. 7, pp. 2435–2443, Jul. 2010.
[34] H.-W. Sim, J.-S. Lee, and K.-B. Lee, "On-line parameter estimation of interior permanent magnet synchronous motor using an extended Kalman filter," *J. Elect. Eng. Technol.*, vol. 9, pp. 600–608, 2014.
[35] Q. Liu and K. Hameyer, "High-performance adaptive torque control for an IPMSM with real-time MTPA operation," *IEEE Trans. Energy Convers.*, vol. 32, no. 2, pp. 571–581, Jun. 2017.
[36] G. Feng, C. Lai, K. Mukherjee, and N. C. Kar, "Current injection-based online parameter and VSI nonlinearity estimation for PMSM drives using current and voltage DC components," *IEEE Trans. Transp. Electrific.*, vol. 2, no. 2, pp. 119–128, Jun. 2016.
[37] Y.-R. Mohamed and T. Lee, "Adaptive self-tuning MTPA vector controller for IPMSM drive system," *IEEE Trans. Energy Convers.*, vol. 21, no. 3, pp. 636–644, Sep. 2006.
[38] R. Ni, D. Xu, G. Wang, L. Ding, G. Zhang, and L. Qu, "Maximum efficiency per ampere control of permanent-magnet synchronous machines," *IEEE Trans. Ind. Electron.*, vol. 62, no. 4, pp. 2135–2143, Apr. 2015.
[39] A. Rabiei, T. Thiringer, M. Alatalo, and E. A. Grunditz, "Improved maximum-torque-per-ampere algorithm accounting for core saturation, cross-coupling effect, and temperature for a PMSM intended for vehicular applications," *IEEE Trans. Transp. Electrific.*, vol. 2, no. 2, pp. 150–159, Jun. 2016.
[40] C. Lee and I. G. Jang, "Topology optimization of the IPMSMs considering both the MTPA and FW controls under the voltage and current limitations," *IEEE Trans. Ind. Electron.*, vol. 70, no. 8, pp. 8244–8253, Aug. 2023.
[41] Q. Dong, B. Wang, L. Xia, Y. Yu, M. Tian, and D. Xu, "A virtual voltage field-weakening scheme of trajectory correction for PMSM model predictive control," *IEEE Trans. Power Electron.*, vol. 38, no. 3, pp. 3044–3056, Mar. 2023.



Kunkun Zuo was born in Xianyang, China, in 1994. He received the B.S. degree in mineral processing engineering from Northeastern University, Shenyang, China, in 2016, and the M.S. degree in control engineering from the Chinese Academy of Sciences, Beijing, China, in 2020. He is currently working toward the Ph.D. degree with Technische Universitaet Muenchen, Munich, Germany.

His research interests include model-predictive control and robust control for electrical drive systems.



Fengxiang Wang (Senior Member, IEEE) was born in Jiujiang, China, in 1982. He received the B.S. degree in electronic engineering and the M.S. degree in automation from Nanchang Hangkong University, Nanchang, China, in 2005 and 2008, respectively, and the Ph.D. degree in electrical engineering from the Institute for Electrical Drive Systems and Power Electronics, Technische Universitaet Muenchen, Munich, Germany, in 2014.

He is currently a Full Professor and the Director of the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Quanzhou, China. His research interests include predictive control and sensorless control for electrical drives and power electronics.

Dr. Wang is an Associate Editor for IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS and IEEE TRANSACTIONS ON ENERGY CONVERSION. As the General Chair, he organized IEEE 5th International Symposium on Predictive Control of Electrical Drives and Power Electronics. He is a Fellow of the Institution of Engineering and Technology.



Zheng Li was born in Tai'an, China, in 1991. He received the B.S. degree in electronic information engineering and the M.S. degree in control engineering from the North University of China, Taiyuan, China, in 2015 and 2018, respectively. He is currently working toward the Ph.D. degree in electrical engineering with the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Quanzhou, China.

His research interests include predictive control and optimal control for electrical drives.



Dongliang Ke (Senior Member, IEEE) was born in Quanzhou, China, in 1986. He received the B.S. degree in electrical engineering and the M.S. degree in power electronics and power drives from Northwestern Polytechnical University, Xi'an, China, in 2010 and 2013, respectively.

He is currently a Postdoctoral Supervisor and Deputy Director of the Electrical Drives Laboratory and a Senior Engineer of the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Quanzhou, China. His research interests include predictive control, flux-oriented control, sensorless control, and identification for electrical drives and power converters.

Dr. Ke was a Session Chair of IEEE 5th and 7th International Symposium on Predictive Control of Electrical Drives and Power Electronics. He is a Member of the Institution of Engineering and Technology and a Senior Member of the Chinese Society for Electrical Engineering.



Ralph Kennel (Senior Member, IEEE) was born in Kaiserslautern, Germany, in 1955. He received the Diploma and Dr.-Ing. (Ph.D.) degrees in electrical engineering from the University of Kaiserslautern, Kaiserslautern, in 1979 and 1984, respectively.

From 1983 to 1999, he was with Robert Bosch GmbH, Gerlingen, Germany. Until 1997, he was responsible for the development of servo drives. From 1994 to 1999, he was a Visiting Professor with the University of Newcastle upon Tyne, Newcastle upon Tyne, U.K. From 1999 to 2008, he was a Professor of Electrical Machines and Drives with Wuppertal University, Wuppertal, Germany. From 2008 to 2021, he was a Professor of Electrical Drive Systems and Power Electronics with the Technical University of Munich, Munich, Germany. His main research interests include sensorless control of ac drives, predictive control of power electronics, and hardware-in-the-loop systems.

Dr. Kennel is a Fellow of the Institution of Engineering and Technology and a Chartered Engineer in the U.K. Within IEEE, he is a Treasurer of the Germany Section.



Marcelo Lobo Heldwein (Senior Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the Federal University of Santa Catarina (UFSC), Florianopolis, Brazil, in 1997 and 1999, respectively, and the Ph.D. degree in electrical engineering from the Swiss Federal Institute of Technology (ETH Zurich), Zurich, Switzerland, in 2007.

He is currently the Head of the Chair of High Power Converter Systems, Department of Energy and Process Engineering, Technical University of Munich, Munich, Germany. From 1999 to 2003, he worked with industry, including R&D activities with the Power Electronics Institute, Brazil and Emerson Network Power, in Brazil and Sweden. From 2007 to 2009, he was a Postdoctoral Fellow with ETH Zurich and UFSC. From 2010 to 2022, he was a Professor with the Department of Electronics and Electrical Engineering, UFSC. His research interests include power electronics, advanced power distribution technologies, and electromagnetic compatibility.

Dr. Heldwein is a Member of the Brazilian Power Electronics Society and the Advisory Board of PCIM Europe.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:52:59 UTC from IEEE Xplore. Restrictions apply.